Finding objects of experiences[1, 2]

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Abstract. This paper develops a semantics of experiential objects, i.e., satisfiable objects associated with experiential episodes that carry satisfaction/accuracy conditions, and uses it to analyze English *find* with small-clause complements. I propose that *find* uniformly introduces an experiential object, and that the embedded predication compositionally specifies the satisfaction conditions of that object within an object-based truthmaker semantics. The account aims to give a unified analysis for the two senses associated with *find*, namely, the object-encounter reading (*John found the door open*) and the evaluative reading (*John found the cake tasty*), without positing two unrelated lexical items or reducing the evaluative use to belief ascription. More generally, the analysis supports a view on which grammar makes direct reference to experiences as objects that can be evaluated for accuracy.

Keywords: find, experiences, truthmaker semantics, experiential objects.

1. Introduction

This study proposes that natural language makes reference to experiences as satisfiable objects, that is, objects that come with satisfaction/accuracy conditions. The empirical focus of investigation concerns *find* with small clauses as exemplified in (1), although more applications of such objects in other domains, such as perception reports, is possible.

(1) a. John found the door open (when he arrived home). (object encounter/discovery)
b. John found the cake tasty (when he tried it). (subjective evaluation)

In the literature on *find* with small clauses, two senses have been identified: one is labeled as the object encounter or discovery sense as in (1a) while the other one has been taken to involve a subjective evaluation for the truth of the embedded sentence as in (1b). The difference in sense has been ascribed to the existence of two lexical items, namely a subjective *find* and an object encounter *find*.

According to some accounts, subjective *find* provides a sense of subjectivity because it has to combine with strictly subjective propositions (however one might want to model this). The issue is then reduced how to define a subjective proposition. Other accounts ascribed subjectivity to the meaning of *find*, that is, these *find* sentences are subjective because they talk about first-hand private experiences of their subjects. Both groups have tried to prevent this subjective *find* from combining predicates like *open* in some way to account for the sense of subjectivity (see §2.2).

Independently, there has also been a discussion about what type of meaning subjective *find* must have. Some analyses treated it as a doxastic predicate on a par with *believe*. Others took *find* to talk about experiences rather than beliefs (see §2.1).

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In this study, I first show that *find* in its subjective use is not a doxastic predicate, but should better be understood as an experiential predicate (§2.1). Second, I argue that subjective *find* having to combine with so-called "subjective" propositions comes with certain empirical issues (§2.2). Third, I provide empirical data to support a unified treatment of *find* for both uses (§2.3). Finally, I provide an object based truthmaker semantics account (Moltmann, 2024) of *find* for both uses, where I introduce experiential objects as results of what is first-hand experienced (i.e., the result of tasting, seeing, hearing, feeling etc.) (§3).

2. Empirical observations

In this section, I provide empirical arguments against the doxastic treatment of subjective *find*, as well as arguments against it having to combine with discretionary propositions.

2.1. Doxastic treatment of *find*

Doxastic analyses adopt the idea that small clause *find* with predicates of personal taste/opinion behaves like a belief report, as in (2), and then add some further constraint (e.g., Stephenson, 2007; Korotkova and Anand, 2021; Kennedy and Willer, 2022). The literature implements the extra constraint in three main ways. One line takes *find* to require that the attitude holder has first-hand experience of the relevant kind (e.g., Stephenson, 2007; Muñoz, 2019; Korotkova and Anand, 2021 among others).[3] A second line requires the embedded clause to count as subjective in the relevant sense (e.g. discretionary or counterstance-contingent content; Coppock, 2018; Kennedy and Willer, 2022). A third connects the effect to judge-sensitivity in the embedded predicate (Lasersohn, 2005; Stojanovic, 2007; Sæbø, 2009; Moltmann, 2010; Pearson, 2012; Lasersohn, 2016 among others).

(2) Harry finds this cake tasty. $\rightsquigarrow$ Harry believes that this cake is tasty.

I will not try to choose among these implementations here. My target is the shared assumption that evaluative *find* has a belief-report core, which I will show is not an accurate characterization of the subjective *find*.

A first contrast concerns what can count as a basis for the attitude. Belief ascriptions readily allow purely report-based justification, while *find* does not, as in (3).

(3) a. John thought/believed/knew that the cake was delicious because a friend told him so.
b. John found the cake delicious #because a friend told him so.

Doxastic approaches try to derive this pattern from a special presupposition attached to evaluative *find*, or from properties of a subjective embedded predicate. But the same first-hand restrictions shows up in object-encounter uses like (4), where no predicate of personal taste is present. This suggests that there is a constraint associated with *find* more generally. If one maintains a two-lexeme split (object-encounter *find* vs. evaluative *find*), then the two *find* end up having the same requirement coincidentally.

[3] Muñoz (2019) assumes a first-hand requirement, but does not analyze *find* itself as doxastic.

(4) John found the vase broken when he came back home, #but he did not see the vase/#because a friend told him so.

A second contrast is that *find*-ascriptions allow explicit denial of the corresponding belief. In (5a), Harry can have an experience that presents the food as bland while declining to endorse that content as a belief (for instance, because he suspects the bruises are distorting the taste). The analogous configuration with a plain belief report is contradictory, hence the infelicity of the continuation in (5b). This is unexpected if *find* simply asserts a belief report.

(5) a. Harry found the food bland because of the bruises on his tongue, but he did not believe/think that it was actually bland, as his favorite cook made it.
b. Harry believed that the food was bland because of the bruises on his tongue, #but he did not believe/think that it was actually bland, as his favorite cook made it.

A related contrast appears in baby-feed contexts. Here *find* can be used to describe the baby's experience on the basis of outward reactions, while a belief predicate is noticeably degraded. The contrast is illustrated in (6).

(6) Context: Harry has a baby of six months old. Whenever he feeds her a particular type of soup, she does not seem to like the taste, and suffers from gas problems. Harry tells his partner:
a. Let's not feed her this soup. She probably finds it sour.
b. Let's not feed her this soup. #She probably believes/thinks that it is sour.

Finally, small clause *find* permits lack of awareness on the part of the finder in a way that belief reports generally resist.[4]

(7) a. John finds Mary amusing though he does not know/realize this yet.
b. #/??John believes that Mary is amusing, though he does not know/realize this yet.

This property of *find* in (7) is shared by perception predicates like *see*.

(8) John sees Mary everyday, but he has not realized it yet/he does not know this yet.

I take these contrasts to argue against treating evaluative *find* as a belief predicate. A better conclusion is that *find* is an experiential predicate like *see* or *hear* (see Hirvonen, 2016).[5]

2.2. Evaluative *find*

A subset of accounts explains the behavior of the evaluative *find* in terms of the subjectivity of its complement. On such accounts, *find* either selects for judge-dependent propositions, or for

[4] There are arguments for dispositional beliefs that need not be consciously accessed (e.g., Rose and Schaffer, 2013). The point here is comparative in that the *not realized* continuation is markedly more natural with *find* than with belief.

[5] See also Muñoz (2019) for arguments that *find* is non-doxastic, developed via contrasts with *consider*. I use *believe* as the comparison point here since *consider* has additional properties that complicate the diagnostics.

a class of discretionary or counterstance-contingent propositions (e.g., Sæbø, 2009; Bouchard, 2013; Coppock, 2018; Kennedy and Willer, 2022 among others).

For the judge based accounts, although particular implementations differ, the general idea is that subjective attitude verbs act as judge shifters. They assign the value of the judge parameter for the embedded clause as the attitude subject, as represented in (9).

(9) $[\![\text{find}]\!] = \lambda p.\lambda x.\ p[\textsc{judge} := x]$

In addition to the commonly discussed issues of treating certain propositions as judge-dependent (see Coppock, 2018 for objections and an alternative outlook-based view; see also Kennedy and Willer, 2022), accounts that separate the evaluative use of *find* from its object-encounter use in this way risk missing a broader generalization, namely that both uses of *find* with small clauses display a first-hand experiential requirement. In accounts that take the two *find*s to correspond to distinct lexical items, this experiential requirement would end up being an accidental shared property.

For example, in object-encounter uses, a small-clause complement strongly favors a direct experiential construal (10a), while clausal complements are more compatible with indirect support ((10b) and (10c)).

(10) *Context:* You and I speak on the phone. You tell me you might have left the front door open. This prompts me to investigate: I open the security app and see that the front-door contact sensor currently reads "open"; when I try to engage the smart lock remotely, the app reports that it cannot lock because the door is open. I never go to the door myself.
a. #After what you told me on the phone, I found the door open.
b. After what you told me on the phone, I found the door to be open.
c. After what you told me on the phone, I found that the door was open.

We observe a parallel complement-sensitivity with evaluative uses, where small clause uses favor direct experience inferences as in (11a), while this requirement disappears with the other complement types (11b)-(11c).

(11) *Context:* You mention that the bakery's new cake might be excellent. I cannot taste it myself, but I consult an official blind-tasting report with published scores and notes, and I also check multiple independent reviews, all converging on the conclusion that the cake is tasty.
a. #After checking the reports, I found the cake tasty.
b. After checking the reports, I found the cake to be tasty.
c. After checking the reports, I found that the cake was tasty.

If the first-hand perception/experience requirement were purely a property of the embedded subjective predicate, two questions arise. First, why does a comparable first-hand requirement also appear with the object-encounter reading of *find*, where there is no such subjective predicate involved? Second, why does the first-hand inference weaken when *find* combines with other clause types, such as infinitivals or *that*-clauses?

If, instead, the first-hand perception/experience requirement with evaluative uses of *find* is treated as a lexical property of *find*, then the parallel requirement in object-encounter uses is expected, provided the two uses are not split into unrelated lexical items. In general, analyses that sharply separate the two constructions into two lexical entries lose the generalization that *find* behaves like other perception/experience reports.

In this sense, a perception verb like *see* also requires the experiencer to have direct perception of what is embedded when the embedded clause is what is labeled a "naked infinitival" in Barwise (1981); Barwise and Perry (1981, 1983), while this requirement disappears with *that*-clauses (e.g., see Saarinen, 1983 and Bayer, 1986).

(12) a. John saw Mary cross the street. (direct perception of the embedded event required)
b. John saw that Mary crossed the street. (indirect perception of the embedded event is allowed)

Kennedy and Willer (2022) recently developed a judge-free rendition of *find* sentences. They do not treat the first-hand experiential requirement of evaluative *find* as a lexical condition on the verb, but derive it from a general "familiarity" condition on subjective attitude reports. They formulate the relevant notion of subjectivity in terms of what they call "counterstance contingency". Somewhat simplifying, a counterstance-contingent proposition is a proposition whose truth cannot be established by worldly facts alone, but requires the attitude holder to take a pragmatic stance on how the relevant facts are to be assessed in context. In order to take such a stance, the attitude holder must have the relevant doxastic access to the facts at issue and, for predicates of personal taste, this access must be grounded in the holder's own affective experience. Accordingly, Kennedy and Willer (2022) derive the first-hand experience inference associated with evaluative *find* from the counterstance-based presupposition that they place on its prejacent.

However, in this account, *find* is still characterized by a condition on a special class of contents that satisfy a counterstance contingent based requirement imposed by evaluative *find*. As already noted at the beginning of this subsection, this leaves the behavior of so-called object-encounter uses unexplained, since those uses display the same first-hand requirement without embedding obviously counterstance-contingent propositions. More generally, in such an account, the role of *find* remains tied to a special kind of subjective/evaluative proposition, rather than to a more general notion of experiential access that could in principle apply also in non-subjective domains.

Crucially, there is an empirical concern that applies to any account on which *find* only combines felicitously with subjective propositions, including judge-shifter analyses. Consider the exhange in (13).

(13) a. A: 2 plus 2 equals 6.
b. B: Mathematicians would not find that accurate/true.
c. # B: Mathematicians would not find 2 plus 2 equal to 6.

Here the previous assertion is evaluated against mathematical facts. The truth of *that is accurate/true* is therefore no more judge-dependent or counterstance contingent than the truth of *two* *plus two equals six*. Yet *find* remains felicitous in (13b), while direct embedding of the mathematical content is infelicitous in (13c). This contrast is unexpected on analyses that explain the distribution of *find* solely in terms of judge-dependence or counterstance contingency of the embedded proposition.[6]

2.3. Evidence for a single *find*

In the following, I present a number of conceptual and empirical pieces of evidence to show that object encounter and evaluative *find* share similarities.

First, in a number of typologically distinct languages, the same lexical item is responsible to provide both senses of *find*. The pattern is too systematic crosslinguistically to suggest that the verb *find* responsible for the discovery sense is accidentally homophonous with the *find* responsible for the subjective evaluation sense. The examples illustrating these readings are presented in (14) and (15).

(14) Evaluative *find*

a. Harry yemeğ-i tuzlu bul-du.
Harry food-ACC salty find-PST
'Harry found the food salty.' (Turkish)

b. Ham molva-şkhimi-s çkva go-taxal-eri
this come.NMZ-POSS.1.SG-LOC more PREVERB-gain.weight-PRTCP
g-dzir-i
2.OBJ-SEE-1.SG.PST
'I found you much fatter this time.'
(Laz, adapted from Bucaklişi et al., 2007)

(15) Object encounter *find*

a. Ev-e gir-dik-ler-inde çocuğ-u baygın bul-du-lar.
house-DAT enter-NMZ-PL-when child-ACC fainted find-PST-PL
'When they came home, they found the child fainted.' (Turkish)

b. Oxori-şa moptisis bere gama-khutsx-eri b-zhir-i.
house-to when.1.SG.CAME child PREVERB-wake.up-PART 1.see-PST
'I found the kid awake when I came home.'
(Laz, adapted from Bucaklişi et al., 2007)

Although object-encounter readings are distinguished from the evaluative readings in the literature, the cross-linguistic systematicity of using *find* in both senses suggests that the two readings

[6] A natural reply from the camp using judge-dependency is to treat predicates such as *accurate/correct/true* as judge-sensitive even in technical domains, thereby making the complement eligible for a judge-shifter analysis. However, unlike paradigmatic predicates of personal taste, these correctness predicates resist overt saturation of a putative judge argument in mathematical cases (e.g. ??*This proof is true/accurate to me / for me*). The fact that dative phrases improve markedly with epistemic matrix predicates (e.g. *This seems accurate to me*) does not by itself support judge-sensitivity of *accurate*. In such cases, the dative presumably realizes the experiencer of *seem*, and the introduced experience provides the evidential basis against which accuracy is assessed, rather than saturating a judge argument of *accurate* itself. In contrast, such a move is not even eligible for counterstance-contingency-based accounts. Positing assessment sensitivity in this case would invite doing so for any proposition, including the one in (13c).

are related to each other.[7]

Second, if there is a discovery *find* different from the subjective one, why is it that when the embedded clause has a subjective predicate, the discovery sense never emerges? These subjective predicates are not incompatible with *discover* (16):

(16) a. John discovered that the cake is tasty.
b. John found the cake tasty. (no discovery sense)

Third, why is it that when the subjective evaluation sense is not available with certain predicates, the discovery sense is also barred? What prevents (17) to be grammatical with a discovery sense?

(17) #John found Paris part of France.

Finally, the discovery and subjective evaluative senses of *find* behave the same in a number of environments. First, both discovery and evaluative *find* imply a defeasible belief (18).

(18) a. John found the door open, but he did not believe/think that it was open.
b. John found the cake tasteless, but he did not believe/think that it was tasteless.

Second, both invoke a defeasible realization inference (19).

(19) a. John found the vase broken, but he did not realize it then.
b. John found Mary amusing, but he did not realize it then.
c. John finds Mary amusing, but he does not know it yet/he does not realize it.

Both object-encounter and evaluative uses of *find* are non-veridical:

(20) a. John found the vase intact, but it was not.
b. John found the cake tasteless, but it was not.

Both object-encounter and evaluative uses of *find* requires first-hand experience with the object in question, when they occur with small clause complements.

(21) a. John found the vase broken, #but he did not see the vase.
b. John found the cake tasteless, #but he did not try it.

Both object-encounter and evaluative *find* mostly prohibits nominal complements:[8]

[7] This does not mean that every language realizes both senses with the same lexical item. In §3, I will propose that *find* can introduce the results of different experiential episodes such as visual, affective, gustatory and so on. This is compatible with the view that languages might assign distinct items for different experiences.

[8] Nominal predicates are not categorically excluded from small-clause uses of *find*. Tom Meadows (p.c.) offers the following illustrative examples:

(i) a. Mary found John a pain in the neck.
b. Mary found John a pleasure to work with.

These suggest that the relative markedness of nominal predicates in small-clause *find* may track the kind of property they express; in particular, whether the property is plausibly available as a directly experiencable target of the first-hand encounter (e.g. affective evaluations like *a pain in the neck*), rather than the syntactic category. By contrast, many sortal/status nominals (e.g. *a soldier*) typically require mediation by cues. One may directly perceive a

(22) a. #John found Mary a soldier. (int: When he saw Mary, Mary was in a soldier state.)
b. #John found Mary a vegetarian. (int: When he saw Mary, John thought that Mary was a vegetarian.)

Finally, both types of *find* eliminate the first hand requirement when they take non-small-clause complements as shown previously in (10)-(11):

Based on the observed data above, I conclude that a unified treatment of *find* is justified. In §3, I will propose a unified account introducing experiential objects with accuracy conditions.[9]

3. Experiential objects and *find*

3.1. Object based truthmaker semantics

The object-based truthmaker semantics has three main features as developed in Moltmann (2024). First, it introduces attitudinal objects into the ontology. Attitudinal objects are the kind of objects such as *beliefs, desires, hopes, fears, or claims* that stand in a satisfaction relation to truthmakers (e.g., states as labeled in Fine, 2017b), similar to how formulas are made true in a state/situation. Accordingly, just as a formula can be verified or falsified by a state in truthmaker semantics, these attitudinal objects can be verified (satisfied) or falsified (violated) by the same states.

Second, the object-based truthmaker semantics takes attitude predicates such as *believe, hope* or *claim* in natural language to stand for complex constructions such as *have the belief*, *have the hope*, or *make the claim*. Hence, attitude predicates like *believe* introduce attitudinal objects of the sorts just mentioned, and light verbs such as *have* or *make* introduce a relation between the attitude holder and these objects.

Finally, embedded clauses, when predicated of attitudinal objects, specify the satisfaction conditions of those objects.

In the object-based truthmaker semantics, attitude predicates introduce attitudinal objects of

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uniform or documentation, but the institutional property itself is ordinarily established inferentially. Consistent with this, infinitival and *that*-clause complements are acceptable in such cases, since they are compatible with discovery via indirect support:

(ii) a. Mary found John to be a soldier.
b. Mary found that John was a soldier.

[9] Vardomskaya (2018) argues for a lexical split between a non-stative "discovery" *find* and a stative evaluative *find*, based on (i) compatibility with the progressive, (ii) pseudo-clefts, and (iii) *for*/*in* adverbials. I do not find these diagnostics decisive. First, the progressive data in (Vardomskaya, 2018: p. 126) are compatible with independently available dynamic construals in both cases (ongoing search/realization for discovery uses; ongoing tasting/evaluation-formation for evaluative uses), and evaluative *find* readily occurs with strongly "now"-oriented adverbials (e.g. *People are finding the new system confusing at the moment*). Second, the pseudo-cleft contrast compares different complement types. When both examples contain a small clause, their acceptability aligns (e.g. ?*What John did was find the treasure missing* vs. ?*What John did was find the soup delicious*). Third, *for*/*in* contrasts track independently motivated factors (notably kind- vs. object-level small-clause subjects and the availability of iterative construals). For example, discovery uses combine naturally with *for* under iteration (e.g. *For weeks, John found the window open whenever he checked*), and evaluative uses can be degraded with *for* when only a single experience is salient (e.g. #*After trying the cake once, Mary found that cake tasty for a year*).

various sorts.[10] The verb *believe* introduces a belief-object, while *claim* introduces a claim-object, and so on. A noun like *belief* denotes a set of belief objects, or its characteristic function. I represent this as in (23).[11]

(23) $[\![\textit{belief}]\!] = \lambda o_e.\, \lambda s_m.\; s \Vdash belief(o)$

Embedded (*that*-)clauses are predicates of attitudinal (and more generally satisfiable) objects specifying their satisfaction/truth conditions. I formalize this by means of an operator EQUIV, which introduces a satisfaction-equivalence relation between a clause and a representational object. Intuitively, EQUIV requires the satisfaction conditions of an attitudinal object to match the truthmaking conditions of the embedded clause. I take clauses to denote characteristic functions of sets of states that exactly verify them as in (24).

(24) $[\![\textit{Mary is a spy}]\!] = \lambda s_m.\; s \Vdash spy(m)$

Here $\Vdash$ is the truthmaking relation between states and formulas only. For representational objects, I use separate predicates for satisfaction as in (25).

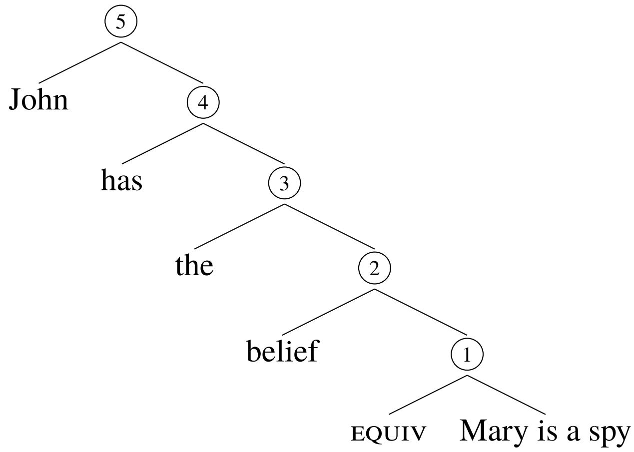
(25) $\mathsf{Sat}(o, s) \iff s$ satisfies o

Given a propositional content $f_{\langle m,t\rangle}$, EQUIV turns f, relative to a predicate P, into a property of satisfiable objects as follows. An object o satisfies $\textsc{equiv}(f)(P)$ in a state s iff (i) o falls under P in s, and (ii) the satisfiers of o are exactly the states that make the clause true.

(26) $[\![\textsc{equiv}]\!] = \lambda f_{\langle m,t\rangle}.\, \lambda P_{\langle e,\langle m,t\rangle\rangle}.\, \lambda o_e.\, \lambda s_m.\, P(o)(s) = 1 \wedge \forall s' \,[\, \mathsf{Sat}(o, s') \iff f(s') = 1\,]$

Consider now an embedded attitude report such as (27).

(27) a. John believes that Mary is a spy.
b.



[10] In the following, I present a typed and compositional version of the object based truthmaker semantics that is not identical to the mechanism provided in Moltmann (2024).

[11] e is the type of entities as usual, m is the type of states (truthmakers). $\Vdash$ is the exact truthmaking relation between states and formulas. For any o and s, $s \Vdash$ belief(o) means s makes $belief(o)$ true and s is wholly relevant to the truth of $belief(o)$, i.e. it does not contain any irrelevant material.

The embedded clause denotes the characteristic function of its verifier set as in (28).

(28) $[\![\text{Mary is a spy}]\!] = \lambda s_m.\ s \Vdash spy(m)$

Given (28), combining EQUIV with the embedded clause clause yields (29).[12]

(29) $[\![①]\!] = [\![\text{EQUIV}]\!]([\![\text{Mary is a spy}]\!])$
$= \lambda P_{\langle e,\langle m,t\rangle\rangle}.\ \lambda o_e.\ \lambda s_m.\ P(o)(s) = 1 \ \wedge\ \forall s' \,[\, \mathsf{Sat}(o,s') \iff s' \Vdash spy(m) \,]$

Applying ① to *belief* results in a predicate of attitudinal objects as in (30).

(30) $[\![②]\!] = [\![①]\!]([\![\text{belief}]\!])$
$= \lambda o_e.\ \lambda s_m.\ s \Vdash belief(o) \ \wedge\ \forall s' \,[\, \mathsf{Sat}(o,s') \iff s' \Vdash spy(m) \,]$

After combining with the embedded clause, EQUIV thus produces a predicate of attitudinal objects that have the right satisfaction conditions with respect to the embedded content. This predicate then becomes the argument of *the*.[13]

(31) $[\![the]\!] = \lambda P_{\langle e,\langle m,t\rangle\rangle}.\ \iota x \,[\, \exists s :\ P(x)(s) = 1]$

The term $\iota x[\exists s :\ P(x)(s) = 1]$ picks out an attitudinal object from the set determined by P. Then, applying *the* to ② in (27b) yields (32).

(32) $[\![③]\!] = [\![the]\!]([\![②]\!])$
$= \iota o[\, \exists s''(s'' \Vdash belief(o) \ \wedge\ \forall s' \,[\, \mathsf{Sat}(o,s') \iff s' \Vdash spy(m) \,]) \,]$

The light verb *has* acts as a relation between the attitude holder and the chosen attitudinal object as in (33).

(33) $[\![\text{has}]\!] = \lambda y_e.\lambda x_e.\lambda s_m.\ s \Vdash have(x,y)$

Combining *has* with ③ and then with *John* gives the denotation in (34a) for the whole clause.

(34) a. $[\![④]\!] = [\![\text{has}]\!]\,[\![③]\!]$
$= \lambda x_e.\lambda s_m.\ s \Vdash have\big(x,\ \iota o[\, \exists s''[\, s'' \Vdash belief(o) \,] \ \wedge\ \forall s' \,[\, \mathsf{Sat}(o,s') \iff s' \Vdash spy(m) \,] \,]\big)$

[12] These objects come also with their violation conditions. In the version developed in Moltmann (2024), the violator set of a belief that S is the set of states falsifying S. In the bilateral truthmaker semantics, a bilateral proposition is taken to be a pair consisting of the verifiers and falsifiers of a clause. Alternatively, in the unilateral version, the falsifiers can be derived from the verifiers based on the relation of exclusion (e.g., Fine, 2017a). In this case, the violators of a representational object can be recast as the excluders of its satisfiers. For reasons of space and clarity, I will abstract away from the violator dimension of such objects in this paper.

[13] Moltmann treats *the* in the relevant constructions as a weak definite as proposed in Srinivas and Legendre (2024). Another possibility is to treat *the* as introducing an ε-term relativized to a contextually supplied choice function (cf. choice-functional treatments of definites/indefinites in Reinhart, 1997; von Heusinger, 1997; Winter, 1997; von Heusinger, 2000, 2004). This would still generate a term that can be anaphorically referred to without the uniqueness presupposition associated with the iota operator. For ease of exposition, I represent *the* as yielding an individual-denoting term built from the predicate contributed by EQUIV and *belief*. This allows the resulting attitudinal object to serve as an argument to predicates like *have* and to be tracked anaphorically in subsequent discourse.

b. $[\![ⓢ]\!] = [\![④]\!]\ [\![\text{John}]\!]$
$= \lambda s_m.\ s \Vdash have\big(j,\ \iota o[\,\exists s''[\,s'' \Vdash belief(o)\,]\ \wedge\ \forall s'\,[\,\mathsf{Sat}(o, s') \iff s' \Vdash spy(m)\,]\,]\big)$

In words, (34b) gives the set of states that verify the sentence *John believes that Mary is a spy*. A state s verifies the sentence iff it verifies that John stands in the *have*-relation to an attitudinal object o picked out by the iota description. This object is required to be a belief, and its satisfiers are exactly the states in which *Mary is a spy* is true. The role of the ι-term here is simply to turn the predicate of such objects into an individual-denoting expression that can serve as an argument to predicates like *have* and can be picked up anaphorically in subsequent discourse. The resulting verifier set can then be predicated of an appropriate object at the matrix level.

3.2. A unified treatment of *find*

I propose that in addition to attitudinal and modal objects, we need experiential objects whose satisfaction conditions represent the accuracy of an experience. The proposal is in the spirit of Moltmann (2024) on attitudinal objects as objects associated with mental/linguistic acts that carry satisfaction conditions. While attitudinal objects were designed primarily for propositional and modal attitudes, experiential objects target the results of experiential episodes (perception, tasting, affective experience, etc.) as they surface in grammar.

In the previous discussions, I argued that *find* cannot be treated as a doxastic predicate, nor as a verb that merely selects subjective propositions. In this subsection, I develop an alternative unified analysis that avoids these problems.

I propose that *find*, in its experiential uses, introduces an experiential object of an appropriate type, depending on what it embeds (visual, gustatory, affective, etc.). This is supported by the fact that we can make explicit reference to such objects, their types, and their accuracy. Consider the exchanges in (35) and (36).

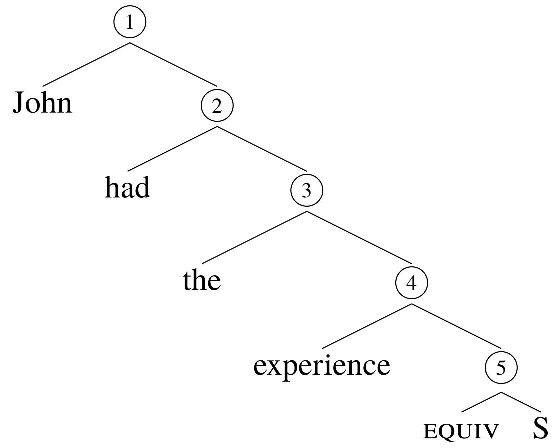
(35) A: John told me that he found the vase intact.
B: I do not think that his visual experience/perception is accurate.

(36) A: John told me that he found the cake tasty.
B: I had the same experience./I think his taste experience is not accurate.

In (35), we can refer to a visual experience; in (36), to a gustatory experience. This suggests that *find* does not merely relate an individual to an abstract proposition, but to an experiential object that can be evaluated for accuracy or inaccuracy. Importantly, the relevant notion of accuracy is not mere extensional truth of the embedded predication. An experience can be accidentally correct (e.g. in hallucination cases) without thereby counting as accurate. What matters is whether the experience is appropriately grounded in (i.e. triggered by) a situation verifying the embedded predication.

Accordingly, I propose that *find* can be decomposed into *have the experience*, as in (37b).

(37) a. John found {the door open, the cake tasty}.

b. 

I take the word *experience* to denote a set of experiential objects (or its characteristic function), parallel to *belief* as shown in (38).

(38) $[\![experience]\!] = \lambda o_e.\, \lambda s_m.\, s \Vdash experience(o)$

As in §3.1, the embedded clause denotes a characteristic function of its verifier set. For illustration, let f_{SC} be the denotation of a small clause (e.g. *the door open*) of type $\langle m, t\rangle$.

In the case of experiential objects introduced by small-clause *find*, I assume a trigger-sensitive variant of EQUIV. Experiential objects come with a triggering relation $\mathsf{Trig}(o, s)$, holding between an experiential object and the directly experienced situation that gives rise to it. The idea is that a verifier counts as satisfying the experiential object only if it can serve as such a first-hand trigger for that object.[14]

(39) $[\![\text{EQUIV}_{\mathsf{Exp}}]\!] = \lambda f_{\langle m,t\rangle}.\, \lambda P_{\langle e,\langle m,t\rangle\rangle}.\, \lambda o_e.\, \lambda s_m.\, P(o)(s) = 1 \wedge \forall s' \,[\, \mathsf{Sat}(o, s') \leftrightarrow [\, f(s') = 1 \wedge \mathsf{Trig}(o, s') \,]\,]$

Applying EQUIV$_{\mathsf{Exp}}$ to f_{SC} and to *experience* yields a predicate of experiential objects whose satisfiers are exactly those verifier states of f_{SC} that also stand in the appropriate triggering relation to the experiential object. This immediately yields the first-hand inference for small-clause *find*. If *find* reports that the subject has an experiential object with a small-clause content, then there must be a first-hand triggering situation verifying that content.

(40) a. John found the door open, #but he did not check the door.
b. John found the cake tasteless, #but he did not try/taste it.

[14] I assume that small-clause *find* introduces experiential objects whose satisfaction conditions are fixed by a trigger-sensitive EQUIV as in (39). It is far from obvious that experiential objects should always be constrained in this way. Different complement types (small clauses vs. infinitivals vs. *that*-clauses) may connect to experiential objects via different mechanisms, and one may want to distinguish more carefully between triggering and satisfaction in those cases.

Furthermore, in this analysis, different senses of *find* arise because of what type of experiential object it can introduce, given what can serve as a first-hand trigger for that type of experience. For example, in (40a), the object-encounter reading arises because the introduced object is a visual experience, for the trigger for that object is the door being open, while in (40b), the introduced object is a taste experience because the trigger for the experience is the cake being tasteless.

Accordingly, we can represent the contribution of *find* in parallel to the attitude case as shown in (41).

(41) $[\![\textit{John found } f_{SC}]\!] = \lambda s_m.\ s \Vdash have\big(j,\ \iota o[\,\exists s''[\,s'' \Vdash experience(o)\,]\ \wedge$
$\forall s'\,[\,\mathsf{Sat}(o,s') \leftrightarrow [\,f_{SC}(s') = 1 \wedge \mathsf{Trig}(o,s')\,]\,]\,]\big)$

In words, (41) says that a state s verifies *John found* f_{SC} iff s verifies that John stands in the *have*-relation to an experiential object o such that (i) o is an experience object and (ii) the satisfiers of o are exactly those states that verify the small-clause content and are eligible as first-hand triggers for o. This captures the first-hand inference without treating *find* as doxastic and without imposing a restriction to subjective complements, while also excluding accidental correctness of the experience in hallucination-like cases.

Finally, one observation is that the experiential requirement of *find* does not disappear under sentential negation as shown in (42).

(42) a. John did not find the food tasty.
b. John did not find the door open.

Both sentences imply that John had the relevant first-hand experience. In (42a), he has tasted the food while in (42b) he has checked the door.

One way to account for this persistence is to build it into the lexical semantics of the *have* relation used in the decomposition of *find*. The assumption is that, for small-clause contents, a positive experiential object with content f comes with a corresponding negative dual whose content is the negated small-clause content. Concretely, let f be a small-clause denotation and let o_f be an experiential object satisfying the $\textsc{equiv}_{\mathsf{Exp}}$-condition for f. We can define its negative dual as in (43).

(43) $-o_f := o_{\neg f}$, where $o_{\neg f}$ is an experiential object satisfying the corresponding $\textsc{equiv}_{\mathsf{Exp}}$-condition for $\neg f$ (and of the same experiential type as o_f).[15]

Thus, the dual is defined by switching the embedded content from f to $\neg f$ while keeping the experiential type fixed. The *have* relation in the decomposition of *find* is then taken to be a partial function with a presupposition that makes the relevant experiential episode available independently of clausal negation (44).

(44) $[\![have_{find}]\!] = \lambda o.\lambda x : have(x,o)\ \vee\ have(x,-o).\ have(x,o)$

[15]This is inspired by negative individuals discussed in Fine (2017a).

$[\![have_{find}]\!](o,x)$ is defined only in contexts where x stands in the underlying *have* relation either to o or to its negative dual $-o$; when defined, it asserts that x has o.

Accordingly, we may schematically represent the positive case as:

(45) [[John found the door open]] $= \lambda s_m.\ s \Vdash have(j, o_f)$ for some o_f satisfying the EQUIV$_{\mathsf{Exp}}$-condition for $f_{door\text{-}open}$

By (44), this presupposes $have(j, o_f)\ \vee\ have(j, -o_f)$. Since $-o_f = o_{\neg f}$, the presupposition says that John stands in the experiential relation either to a door-open experiential object or to its door-not-open counterpart. Sentential negation yields:

(46) [[John did not find the door open]] $= \lambda s_m.\ s \Vdash \neg have(j, o_f)$, with the presupposition $have(j, o_f)\ \vee\ have(j, o_{\neg f})$

Because the presupposition is unaffected by clausal negation, (42b) still requires John to stand in the *have* relation to a relevant experiential object. When combined with the negated assertion, the standing interpretation is that John has $o_{\neg f}$ rather than o_f. This accounts for the intuition that (42b) is infelicitous if John never checked the door at all. The same reasoning extends to evaluative small clauses such as *John did not find the food tasty*.

In general, we proposed that *find* does not predicate directly over propositions or situation types, but over experiential objects. Experiential objects are content-bearing entities that carry satisfaction condition. In small-clause uses, the embedded predication fixes those satisfaction conditions via EQUIV$_{\mathsf{Exp}}$, and the trigger-sensitivity of that operator ensures that accuracy is evaluated relative to states that can serve as first-hand triggers for the reported experience.

4. Conclusion

This paper argued that English *find* with small clauses is best analyzed as an experiential predicate, rather than as a doxastic attitude verb or as a verb that merely selects for subjective complements. I proposed a unified account in an object-based truthmaker semantics (Moltmann, 2024) on which *find* introduces an experiential object whose satisfaction (accuracy) conditions are determined by the embedded predication via a trigger-sensitive composition (i.e. satisfaction is restricted to verifier states that can serve as first-hand triggers for the relevant experience).

Framing *find* in terms of experiential objects yields several advantages over the alternatives discussed in §2. First, it captures the non-doxastic behavior of evaluative *find*. Since *find* talks about experiential objects, a *find*-ascription does not entail endorsement of the embedded content of the experience as a belief object.

Second, the analysis does not require a restriction to counterstance-contingent or judge-dependent propositions. Instead, the first-hand inference associated with small-clause embeddings follows from the nature of experiential objects and from the way small-clause contents determine their satisfaction conditions (and, under the simplifying assumption adopted here, constrain what can count as a first-hand trigger).

Third, the account supports a unified lexical treatment of *find*. In particular, the same compositional mechanism applies to both object-encounter and evaluative cases. The apparent difference in sense arises because different small-clause situations naturally trigger different types of experiential episodes and thus yield different kinds of experiential objects (visual, gustatory, affective, etc.), rather than because there are two distinct lexical items. Of course, there may also be lexical constraints on which experiential objects *find* can introduce, but compared to verbs like *see* and *hear*, *find* appears relatively underspecified with respect to experiential type as suggested in Hirvonen (2016).

A further outcome of the present view is a constraint on when *find* can combine with small clauses at all. Since *find* introduces an experiential object and the small clause fixes its accuracy conditions in a trigger-sensitive way, the analysis predicts that small-clause *find* should be infelicitous whenever the embedded predication does not describe a situation that can plausibly serve as a first-hand trigger of the relevant kind of experience. This helps explain why certain contents are excluded under *find* in any small-clause use (neither a discovery nor an evaluation reading is available) in cases such as #*John found Paris part of France*, #*John found* $2+2$ *equal to* 4, or #*John found the house wooden*.[16]

More generally, the proposal supports a broader view on which grammar can refer to experiences as satisfiable objects that are assessable for accuracy. While this paper focused on *find*, the same perspective suggests a way of treating other experience reports (e.g. perception predicates) within a uniform object-based truthmaker semantics framework.

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[16]One can, of course, experience perceptual cues relative to woodenness (e.g., texture, color, smell), but woodenness itself as such is not a perceptible state. The predication typically involves an additional classificatory step that goes beyond what is directly experienced.

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